

Prompt Engineering

Programming AI with

Everyday Language

Just like we program computers with code, we program AI with language. Better prompts give better answers — and that's a skill anyone can master.



What Is Prompt Engineering?



Prompt engineering is the process of **writing, refining, and improving instructions** given to an AI model to get accurate and useful answers.

❌ Bad Prompt

"Tell me about Python." → Generic, unfocused output

✓ Good Prompt

"Explain Python to a first-year engineering student. Include variables, loops, functions. Limit to 300 words."
→ Structured, clear output

The 4 Parts of a Good Prompt

Every effective prompt contains four key ingredients. Include all four and the AI knows exactly what you want — no guessing required.

1

Instruction

What the AI should do. *"Summarize this article"* or *"Write a Python function"*

2

Context

Background details that shape the response. *"You are an HR recruiter"*

3

Input Data

The actual information to process — a resume, text block, or dataset.

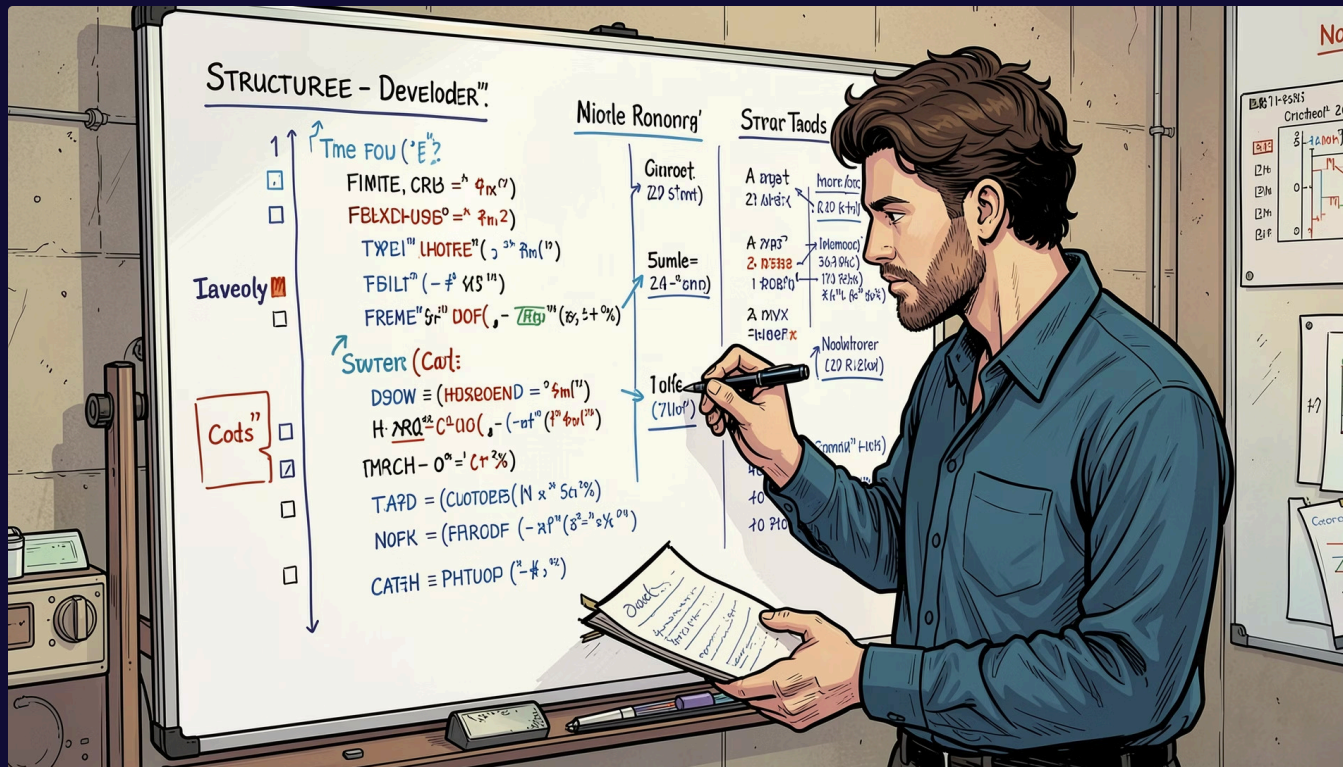
4

Output Format

How the answer should look. *"Return in JSON"* or *"Put it in a table"*

Structured Prompts

Beginners write vague sentences.
Professionals use a clear layout to get exact results without the AI guessing the audience or format.



Example Structured Prompt

Role: You are a senior Python developer.
Task: Write a program to reverse a string.
Constraints: Use only functions.
Output: Provide code and an explanation.

Common Prompting Types



Role Prompting

Assign the AI a personality. *"Act as a doctor"* — changes how it speaks and thinks.



Zero-Shot

Ask directly with no examples. *"Translate: Hello"*



Few-Shot

Give a few examples first so the AI learns the pattern before answering.



Chain of Thought

Ask the AI to reason step-by-step. Ideal for math and logic puzzles.

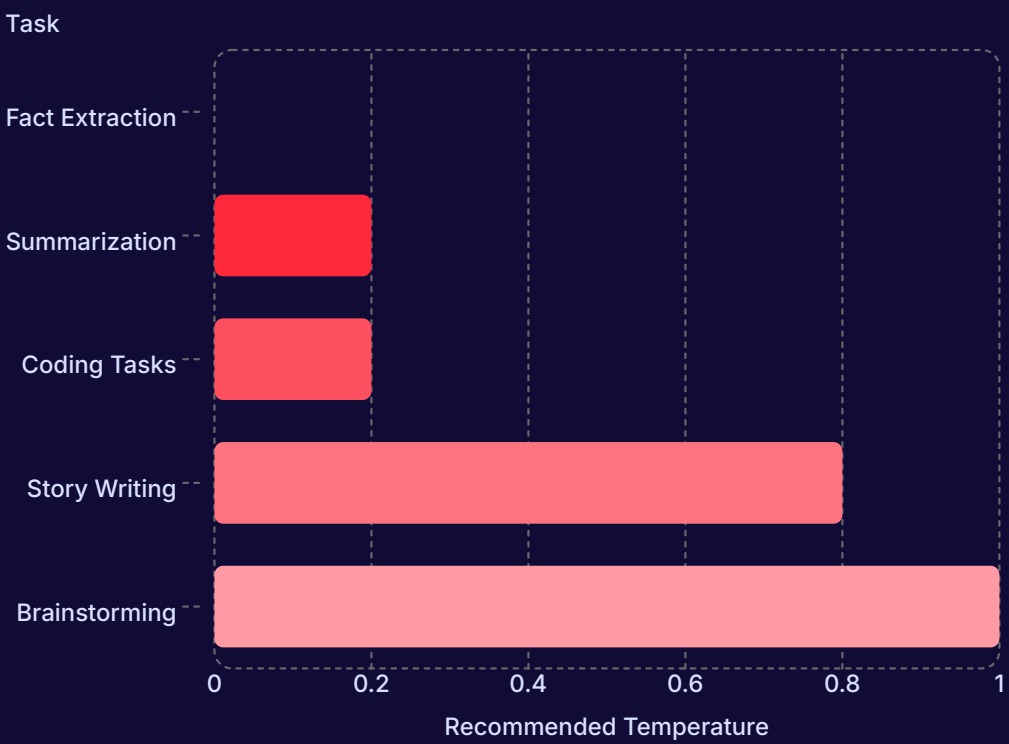
Prompt Engineering Best Practices

- **Be Specific**
Instead of "Tell me something," say *"Explain machine learning to a beginner."*
- **Give Clear Instructions**
Instead of "Write code," say *"Write a Python function to find factorial recursively."*
- **Break Down Complex Tasks**
Ask for Step 1, Step 2, Step 3 — don't dump everything in one vague request.
- **Define Output Format**
Request JSON, markdown tables, or bullet lists for consistent, usable results.

LLM Parameters: Temperature

Temperature controls how random or creative the AI is — from 0 (predictable) to 2 (highly creative). The right value depends on your task.

Task	Recommended Temp
Fact Extraction / Auditing	0.0
Summarization / Math	0.2
Coding Tasks	0.1 – 0.3
Story Writing	0.8
Brainstorming Ideas	1.0



More LLM Parameters

Max Output Tokens

Sets the maximum answer length. Too small and responses get cut off mid-sentence.

Top-P (Nucleus Sampling)

The AI samples from top words whose probabilities sum to P. Low = focused; High = creative.

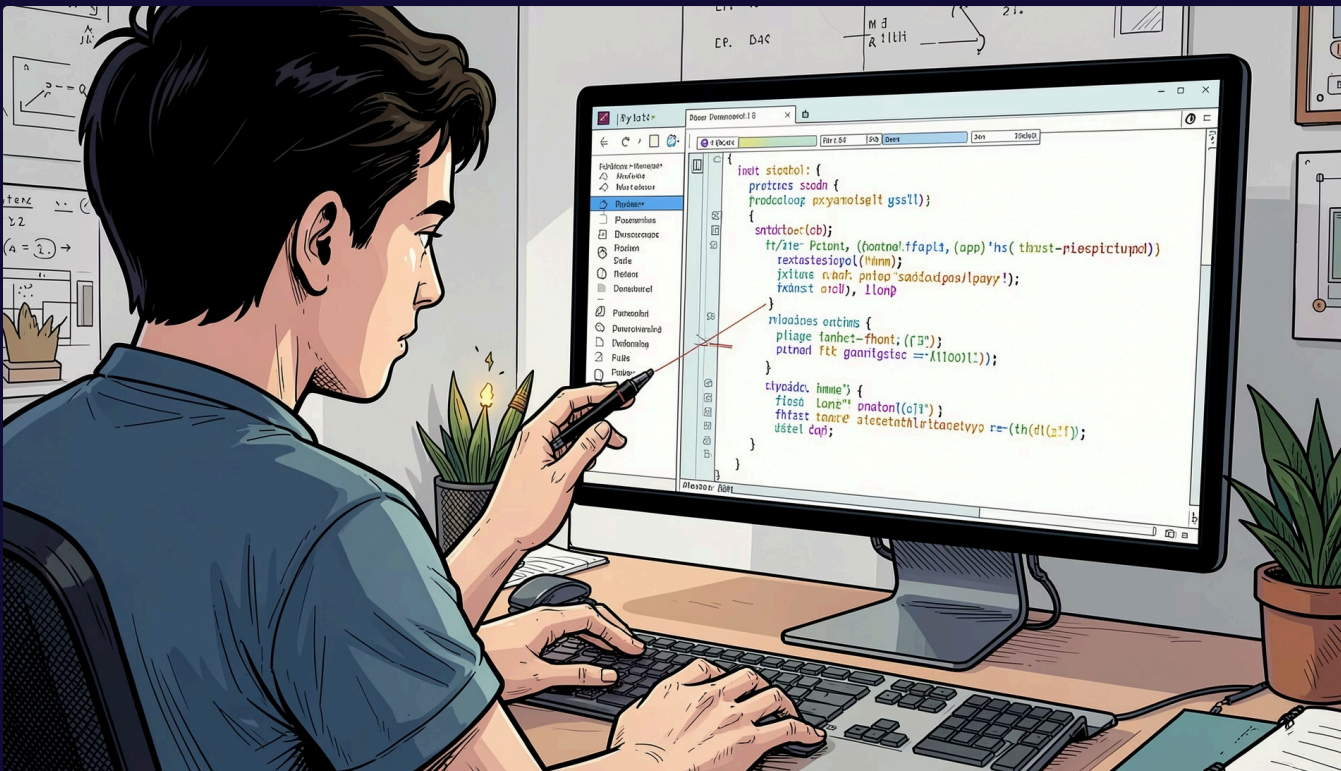
Top-K

The AI only considers the top K most likely next words. *Top-K = 3* means it picks from only the 3 best options.

Seed

Ensures reproducibility. Same prompt + same seed = **identical output every time**. Great for testing.

Generating Code with Prompts



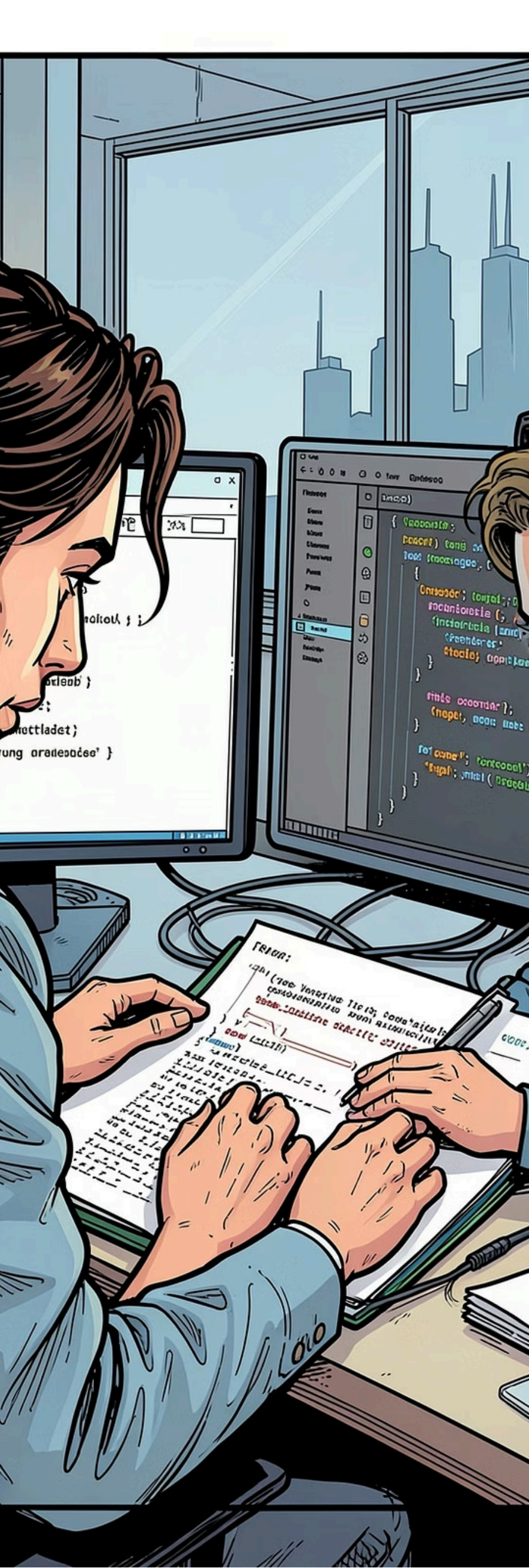
Companies use prompt engineering to auto-generate code safely. **Always provide clear requirements** instead of vague requests.

⚠️ ❌ Bad Prompt

"Write code." — No language, no task, no constraints.

✅ ✓ Good Prompt

"Write a Python function to check whether a string is a palindrome. Use functions, handle uppercase, $O(n)$ complexity. Explain each line."



ADVANCED TECHNIQUE

Advanced Code Prompt Template

Combine role, task, constraints, and output format for production-ready code generation.

Advanced Code Prompt

You are a senior software engineer.

Task: Create a Python CLI application.

Requirements:

- Read CSV file, calculate average salary
- Handle exceptions, follow PEP8

Output: Complete code, explanation, time complexity, and test cases.

Role

Assign expertise level

Task

Define the goal clearly

Constraints

Set rules and standards

Output

Specify deliverables